An explicit local zero-count bound for the Riemann heat deformation

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Abstract

Polymath established local zero counts of order $\log X$ for the Riemann heat deformation, uniformly for positive heat times at most $1/2$. We extract one explicit bound on a restricted height domain. In the physical coordinate $H_0(z) = \xi((1+iz)/2)/8$, the number of zeros with $X < \operatorname{Re} z \le X+1$, counted with multiplicity, is at most $(6/5)\log(X+2)+22$ for $0 < t \le 1/2$ and $X \ge 2 \cdot 10^6$. The proof makes an off-axis comparison uniform in time, extends a Riemann–Siegel remainder estimate to the two real-part ranges needed there, and applies a positive logarithmic-derivative sum. A separate fixed-index remainder is retained in the truncation estimate. All numerical constants and the cutoff are evaluated. The lower-height range, a cumulative error constant and an application-specific gain are not supplied. This draft has not undergone round review or specialist human review, and makes no claim about the Riemann hypothesis or an improved de Bruijn–Newman bound.

Draft status. This manuscript assembles the two research rounds of 17 October 2026 at the user's request, before the deferred review. The derivations and finite arithmetic checks used `gpt-6.1-sol` (Codex), reasoning effort `ultra`, with same-model development assistance. Compilation and authoring checks do not constitute independent mathematical review. No literature-priority claim is made. The author field identifies the intended recipient of the draft, rather than asserting his endorsement.

1 Main result and relation to existing estimates

Let

$$\Phi(u) = \sum_{n\ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1.1}$$

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, \mathrm{d}u. \tag{1.2}$$

The normalization is

$$H_0(z) = \tfrac{1}{8}\xi((1+iz)/2), \qquad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \partial_t H_t = -\partial_z^2 H_t. \tag{1.3}$$

Thus a zeta ordinate of magnitude T corresponds to physical coordinate $x = 2T$. Every logarithm without a capital Log is natural. For $X \ge 0$, define

$$N_t(X) = \sum_{0<\operatorname{Re}\rho\le X} \operatorname{mult}(\rho), \qquad K_t(X) = N_t(X+1) - N_t(X), \tag{1.4}$$

where the sum is over zeros of H_t. A zero on the right wall of a window is included, one on its left wall is excluded, and a real double zero contributes two.

Theorem 1.1 (Explicit unit-window bound)**.** *For $0 < t \le 1/2$ and $X \ge 2 \cdot 10^6$,*

$$K_t(X) \le \frac{82123}{72000}\log(X+2) + \frac{574}{27} \le \frac{6}{5}\log(X+2) + 22. \tag{1.5}$$

The count includes complex and multiple zeros; no reality, simplicity or separation hypothesis is imposed.

Theorem 1.5(iv) of Polymath [1] already establishes a time-uniform estimate $K_t(X) = O(\log(2+X))$, together with a cumulative estimate. The purpose here is to evaluate constants for one local version on a declared domain, rather than to establish the first asymptotic. The cutoff is conservative and is not claimed optimal. The range $200 \le X < 2 \cdot 10^6$ remains outside the present bound. No consumer has yet supplied a numerical tolerance against which to measure an application gain.

The analytic input to Theorem 1.1 is the following comparison. The nonvanishing carrier M_t is defined in Section 2.

Theorem 1.2 (Uniform off-axis comparison)**.** *If $0 < t \le 1/2$, $x \ge 2 \cdot 10^6$ and $3 \le y \le 4$, then*

$$\left| \frac{H_t(x+iy)}{M_t((1+y-ix)/2)} - 1 \right| < \frac{96624791}{147000000} < \frac{33}{50} < \frac{7}{10}. \tag{1.6}$$

The proof of Theorem 1.2 appears in Section 6. Section 7 then proves Theorem 1.1 by Cauchy's estimate and a positive zero sum. The argument needs no compactness constant and no finite zero survey.

2 Carriers and imported analytic estimates

On $\mathbb{C} \setminus (-\infty, 1]$, with principal logarithms, set

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp\left\{\left(\frac{s}{2} - \frac{1}{2}\right)\operatorname{Log}\frac{s}{2} - \frac{s}{2}\right\}, \tag{2.1}$$

$$\alpha(s) = \frac{M_0'(s)}{M_0(s)} = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2}\operatorname{Log}\frac{s}{2\pi}, \tag{2.2}$$

$$M_t(s) = e^{t\alpha(s)^2/4}M_0(s), \qquad \frac{M_t'}{M_t}(s) = \alpha(s) + \frac{t}{2}\alpha(s)\alpha'(s). \tag{2.3}$$

The functions have the expected conjugation symmetry and are nonzero on the domains used below. We use the conventions

$$T = \frac{x}{2}, \quad T' = T + \frac{\pi t}{8}, \quad Q = \frac{x}{4\pi}, \quad L = \log Q, \quad a = \sqrt{\frac{T'}{2\pi}}, \quad N = \lfloor a \rfloor, \tag{2.4}$$

and

$$s_+ = \frac{1+y-ix}{2}, \qquad s_- = \frac{1-y+ix}{2}, \qquad b_n^t = e^{t\log^2 n/4}, \qquad B_t(z) = M_t((1-iz)/2). \tag{2.5}$$

For $3 \le y \le 4$, the real parts of $s_\pm$ belong to $[-3/2, 5/2]$. Note that $a^2 = Q + t/16$.

We state the imported ingredients precisely. All equation and theorem numbers attributed to Polymath refer to arXiv version 2 of [1]. First, H_t is even and entire of order one, $H_t(iy) > 0$ for real y, and its zeros lie in $|\operatorname{Im} z| \le 1$ for $0 \le t \le 1/2$; see its Section 3 and Theorem 3.2.

To describe its expansion (39), write

$$c(s) = \frac{s(s-1)}{16}\pi^{-s/2}\Gamma(s/2), \qquad r_{0,n}(s) = c(s)n^{-s},$$

and

$$R_{0,N}(s) = c(s)\int_{N\swarrow N+1} \frac{w^{-s}e^{i\pi w^2}}{e^{\pi i w} - e^{-\pi i w}}\,\mathrm{d}w.$$

The contour has direction $e^{5\pi i/4}$ and crosses $(N, N+1)$. For nonreal s, define $r_{t,n}$ and $R_{t,N}$ by Gaussian convolution in s, for example

$$R_{t,N}(s) = \frac{1}{\sqrt{\pi}}\int_{\mathbb{R}} R_{0,N}(s + \sqrt{t}\,v)e^{-v^2}\,\mathrm{d}v. \tag{2.6}$$

For $F^*(s) = \overline{F(\overline{s})}$, the exact formula is

$$\begin{aligned} H_t(z) = {} & \sum_{n=1}^{N} r_{t,n}(s_-) + \sum_{n=1}^{N} r^*_{t,n}(s_+) \\ & + R_{t,N}(s_-) + R^*_{t,N}(s_+). \end{aligned} \tag{2.7}$$

This formula and its contour shifts are imported from [1, (39)–(41)]; sub-Gaussian growth on the shifted lines justifies the Gaussian integrals.

Proposition 6.1 of [1] applies for every real σ, $T > 10$, $n \ge 1$ and $0 < t \le 1/2$. It gives

$$r_{t,n}(s) = M_t(s)b_n^t n^{-s-(t/2)\alpha(s)}(1 + \theta_{t,n}(s)), \quad s = \sigma + iT, \tag{2.8}$$

$$|\theta_{t,n}(s)| \le \exp\left(\frac{(t^2/8)|\alpha(s) - \log n|^2 + t/4 + 1/6}{T - 3.33}\right) - 1. \tag{2.9}$$

The r functions have conjugation symmetry, so the same estimates hold at the reflected arguments. We also import the general-real-part coefficient and remainder bounds of Proposition 6.2, including $|C_0(p)| \le 1/2$, and the explicit Stirling estimate of Lemma 5.1(v). We do not extend the final constants of its Proposition 6.3 beyond their stated real-part range. Section 4 rebuilds the needed remainder estimate instead.

3 Bounds for the ordinary terms

Lemma 3.1 (Heat weights and termwise error)**.** *For $x \ge 2 \cdot 10^6$, $0 < t \le 1/2$, $-3/2 \le \sigma \le 5/2$ and $s = \sigma \pm iT$, uniformly for $1 \le n \le N$,*

$$\operatorname{Re}\alpha(s) \ge L/2 - 13/x^2, \tag{3.1}$$

$$|\alpha(s) - L/2 \mp i\pi/4| \le \frac{11}{2x} + \frac{25}{4x^2} < \frac{6}{x}, \tag{3.2}$$

$$W_n(s) := b_n^t n^{-(t/2)\operatorname{Re}\alpha(s)} \le 1, \qquad |\theta_{t,n}(s)| < \frac{1}{50000}. \tag{3.3}$$

Proof. In (2.2), the real rational terms are bounded below by $-13/x^2$, and the log-modulus correction to $L/2$ is nonnegative. The inequalities $|\arctan(\sigma/T)| \le |\sigma|/T$ and $\log(1+\sigma^2/T^2) \le \sigma^2/T^2$, together with the rational terms, prove (3.2). Since $a^2 = Q + t/16$,

$$\log N \le L/2 + \frac{\pi}{16x}, \qquad |\alpha(s) - \log n| \le L/2 + 1 \le \log x.$$

The exact identity

$$W_n(s) = \exp\left\{\frac{t}{4}\log n(\log n - 2\,\mathrm{Re}\,\alpha(s))\right\} \tag{3.4}$$

and the bound $2\,\mathrm{Re}\,\alpha(s) - \log N \ge L/2 - \pi/(16x) - 26/x^2 > 0$ prove $W_n \le 1$. The heat coefficient in (3.4) is $t/4$.

Because $T - 3.33 \ge x/3$, the exponent in (2.9) is at most

$$u(x) = \frac{3(\log x)^2/32 + 7/8}{x}.$$

Both $(\log x)^2/x$ and $1/x$ decrease here. At the left endpoint, $\log(2\cdot 10^6) < 15$, and therefore

$$u(x) \le \frac{703}{32\cdot 2\cdot 10^6} < \frac{1}{8}, \qquad e^{u(x)} - 1 \le \frac{8}{7}u(x) < \frac{1}{50000}.$$

This proves the termwise error assertion. □

The dominant branch has real part $(1+y)/2 \ge 2$. Its leading normalized approximant is exactly 1, and its other terms total at most

$$\sum_{n=2}^{N} n^{-(1+y)/2} W_n(s_+) \le \zeta(2) - 1 \le \frac{95}{147}. \tag{3.5}$$

Here and below $\zeta(2) \le (22/7)^2/6 = 242/147$ is used as a convenient rational upper bound.

4 A remainder estimate for both real-part ranges

Lemma 4.1 (A remainder bound on an enlarged real-part interval)**.** *Let*

$$T \ge 10^6, \qquad -\frac{3}{2} \le \sigma \le \frac{5}{2}, \qquad 0 < t \le \frac{1}{2},$$

and set

$$T' = T + \frac{\pi t}{8}, \qquad a = \sqrt{\frac{T'}{2\pi}}, \qquad N = \lfloor a \rfloor.$$

For the remainder $R_{t,N}$ defined in [1, equations (38)–(41)], one has

$$\left|R_{t,N}(\sigma + iT)\right| \le \exp\left(\frac{t\pi^2}{64}\right)\left|M_0(iT')\right|. \tag{4.1}$$

Proof. We use the general-real-part estimates of [1, Proposition 6.2], rather than extending the stated domain of its Proposition 6.3. Write

$$D = T - 3, \qquad p = 1 - 2(a - N), \qquad u = \sigma + \sqrt{t}\,v.$$

Expectation below refers to the density

$$\frac{1}{\sqrt{\pi t}}\exp\!\left(-\frac{(u-\sigma)^2}{t}\right)\,du.$$

In particular, $a > 300$ throughout the proof.

The Gaussian envelope. Use the shift $\beta = \pi i/4$ in [1, equation (41)]. Its imaginary displacement is $\pi t/8$, so both the original and shifted contours remain in the upper half-plane. Apply Proposition 6.2 to $u + iT'$. After extracting the common carrier and its unit-modulus phase, the integrand is

$$e^{\eta(u)}\left(C_0(p) + \sum_{k=1}^{K_u}\frac{C_k(p,u)}{a^k} + RS_{K_u}(u+iT')\right),$$

where K_u will be chosen below. The extracted carrier has modulus

$$\exp\!\left(\frac{t\pi^2}{64}\right)|M_0(iT')|.$$

The explicit Stirling estimate in [1, Lemma 5.1(v)], the formula for α in its equation (9), and the derivative bound in its equation (43) give

$$\begin{aligned}|\eta(u)| &\le \frac{u^2}{4(T'-3)} + \frac{3|u|}{2T'} + \frac{1}{6(T'-0.66)}\\ &\le \frac{u^2+6|u|+2/3}{4D} \le \frac{u^2+1}{D}.\end{aligned} \tag{4.2}$$

Indeed, the horizontal Taylor segment from iT' to $u + iT'$ has constant imaginary part T', and $6|u| \le 3u^2 + 3$. Thus this estimate is valid for every real u; no boundedness assumption on u enters the Taylor argument.

For a real number b, put

$$J_b = \mathbb{E}\exp\!\left(\frac{u^2+1}{D} + bu\right), \qquad h = 1 - \frac{t}{D}.$$

Completing the square gives the exact formula

$$J_b = h^{-1/2}\exp\!\left(\frac{\sigma^2+1}{D} + b\sigma + \frac{t(\sigma/D + b/2)^2}{h}\right). \tag{4.3}$$

The simpler inequality $u^2 \le 2\sigma^2 + 2tv^2$ also gives

$$J_0 \le \exp\!\left(\frac{2\sigma^2+t+1}{T-6}\right) \le \exp\!\left(\frac{14}{T-6}\right) < \frac{4}{3}. \tag{4.4}$$

For $b = \log 9$ or $b = -\log 2$, the factor in (4.3) remaining after removal of $e^{b\sigma}$ is less than 2. To see this, use

$$h \ge \frac{999}{1000}, \qquad \frac{\sigma^2+1}{D} < \frac{1}{1000}, \qquad \frac{|\sigma|}{D} < \frac{1}{1000}, \qquad -\frac{1}{2}\log h < \frac{1}{1000}.$$

Since $\log 3 < 11/10$ and $\log\sqrt{2} < \log 3$, its logarithm is less than

$$\frac{2}{1000} + \frac{1}{2}\left(\frac{1101}{1000}\right)^2\frac{1000}{999} < \frac{2}{3} < \log 2.$$

Coefficient summation and the fixed-index remainder. Choose

$$m = \lfloor 2a^2 \rfloor, \qquad K_u = \begin{cases} 1, & u \geq 0, \\ \max(\lfloor -u \rfloor + 3, m), & u < 0. \end{cases}$$

When $u < 0$, this choice satisfies $K_u + u \geq 2$, as required by [1, equation (57)]. Define

$$S(u) = \sum_{k=1}^{K_u} \frac{|C_k(p,u)|}{a^k} + |RS_{K_u}(u + iT')|, \qquad E_a = a^{-1}e^{-a^2/2}.$$

The following pointwise bound includes every remainder term:

$$\begin{aligned} S(u) \leq \frac{0.2\,9^u}{a - 0.865} + \frac{0.029\,2^{-u}}{a - 0.353} \\ + \frac{2^{-u}}{2} \sum_{2a^2 \leq k \leq -u+4} \frac{1.1^k \Gamma(k/2)}{a^k} + E_a. \end{aligned} \tag{4.5}$$

All sums over k have integer indices; an empty sum is zero.

Here are the details of the summation. For $u \geq 0$, the bounds in [1, equations (54) and (56)] give

$$S(u) \leq \frac{0.200\,9^u}{a} + \frac{0.173\,2^{3u/2}}{a^2} \leq \frac{0.2\,9^u}{a - 0.865}.$$

The value $u = 0$ can be assigned by continuity and has no effect on the integrals. For $u < 0$, the coefficients with $k \leq m$ satisfy

$$\sum_{k=1}^{m} \frac{|C_k(p,u)|}{a^k} \leq 0.036\,2^{-u} \left(\frac{0.445\sqrt{\pi}}{a} + \frac{0.247}{a^2} + \frac{0.098}{a^3} \right) \leq \frac{0.029\,2^{-u}}{a - 0.353}.$$

This follows from [1, equation (55)] by separating even and odd indices and using

$$\frac{\Gamma((k+2)/2)}{a^{k+2}} = \frac{k}{2a^2} \frac{\Gamma(k/2)}{a^k} \leq \frac{\Gamma(k/2)}{a^k} \qquad (k \leq 2a^2).$$

The constants bound the corresponding geometric sums.

If $K_u > m$, every coefficient beyond m has index greater than $2a^2$. Moreover the remainder index satisfies

$$K_u + 1 = \lfloor -u \rfloor + 4 > 2a^2, \qquad K_u + 1 \leq -u + 4.$$

The bounds in equations (55) and (57) of [1] therefore place these additional coefficients and the remainder inside the tail sum in (4.5).

If $K_u = m$, the remainder need not belong to that tail. To bound it, put $\nu = (m+1)/2$, so that $a^2 < \nu \leq a^2 + 1/2$. Equation (57) and explicit real Stirling give

$$\begin{aligned} |RS_m(u + iT')| &\leq \frac{1}{2}\Gamma(\nu)(1.1/a)^{2\nu} \\ &\leq \frac{3}{2a} \exp\left(\nu \log(\nu/a^2) - \frac{4}{5}\nu \right) \\ &< \frac{5}{a} e^{-4a^2/5} \leq E_a. \end{aligned}$$

We used $\Gamma(\nu) \le 3\nu^{\nu-1/2}e^{-\nu}$, $\log(1.21) < 1/5$, $\log(\nu/a^2) \le 1/(2a^2)$, and $3a^2/10 > \log 5$. Thus the added E_a accounts for the fixed-index remainder even when the displayed tail range is empty. This proves (4.5) in every case.

Integration of the four contributions. Let $\delta_1, \dots, \delta_4$ denote the expectations of the four terms in (4.5), each multiplied by $\exp((u^2+1)/D)$. The moment bounds above give

$$\begin{aligned}\delta_1 + \delta_2 &< \frac{0.4\,9^\sigma + 0.058\,2^{-\sigma}}{a - 0.865} \\ &\le \frac{97.374}{a - 0.865} \le \frac{98}{a}.\end{aligned} \tag{4.6}$$

Here $9^\sigma \le 243$ and $2^{-\sigma} < 3$.

In the third contribution, enlarge the integer summation to $k \ge 14$. Its integration region has $u \le 4 - k \le -10$. Since $|\sigma| \le 5/2$,

$$(u-\sigma)^2 \ge \frac{9}{16}u^2, \qquad \frac{u^2+1}{D} - \frac{(u-\sigma)^2}{t} \le \frac{1}{D} - \frac{u^2}{2t}.$$

Furthermore,

$$\int_{-\infty}^{4-k} e^{-u^2/(2t) - u\log 2}\,du \le \frac{2t}{k-6} e^{-(k-4)^2 + (k-4)\log 2}.$$

For

$$f(k) = 1.1^k \Gamma(k/2) e^{-(k-4)^2 + (k-4)\log 2},$$

one has $f(14), f(15) < 10^{-30}$ and

$$\frac{f(k+2)}{f(k)} = 2.42k\,e^{-4k+12} < 1 \qquad (k \ge 14).$$

The initial bounds follow, for example, from $\Gamma(7) = 720$, $\Gamma(15/2) = 135135\sqrt{\pi}/128$, $\sqrt{\pi} < 2$, and $e > 8/3$. Summing the resulting geometric series in $1/a$ gives

$$\delta_3 \le \frac{4 \cdot 10^{-30}}{a^{14}} < \frac{1}{a}. \tag{4.7}$$

The fourth contribution satisfies

$$\delta_4 = E_a J_0 < \frac{4}{3} E_a < \frac{1}{a}. \tag{4.8}$$

Finally, [1, equation (58)] gives $|C_0(p)| \le 1/2$. Taking absolute values in the shifted integral and using (4.2) therefore yields

$$\frac{|R_{t,N}(\sigma + iT)|}{e^{t\pi^2/64}|M_0(iT')|} \le \frac{1}{2}J_0 + \sum_{j=1}^{4}\delta_j < \frac{1}{2}\cdot\frac{4}{3} + \frac{100}{300} = 1.$$

This proves (4.1) with all constants uniform on the stated domain. □

5 Ratios of the carriers

Lemma 5.1 (Carrier comparisons)**.** *On the domain of Theorem 1.2,*

$$\left|\frac{M_t(s_-)}{M_t(s_+)}\right| < 2Q^{-y/2}, \tag{5.1}$$

$$e^{t\pi^2/64}\frac{|M_0(iT')|}{|M_t(s_+)|} \le Q^{-(1+y)/4}. \tag{5.2}$$

Proof. Put $A = (1+y)/2 \in [2, 5/2]$ and $C = 1 - A \in [-3/2, -1]$. For $-3/2 \le \sigma \le 5/2$, direct use of (2.2) gives

$$|\operatorname{Re}\alpha(\sigma+iT) - L/2| \le 85/(16T^2), \qquad |\operatorname{Im}\alpha(\sigma+iT) - \pi/4| \le 11/(4T),$$
$$|\alpha(\sigma+iT)| \le L/2+1, \qquad |\alpha'(\sigma+iT)| \le 1/(2T) + 3/(2T^2).$$

Integrating the real part of the logarithmic derivative in (2.3) from $C+iT$ to $A+iT$ gives

$$\log\left|\frac{M_t(s_-)}{M_t(s_+)}\right| \le -\frac{y}{2}L + \frac{85}{x^2} + (L/2+1)(1/x + 6/x^2) \le -\frac{y}{2}L + \frac{\log x}{x}.$$

For example, the final inequality follows from $L \le \log x$, $85/x \le 1/100$, $6/x \le 1/100$ and $\log x \ge 3$. Conjugation equates the denominator's modulus to that at $A+iT$. Since $\log x/x < 1/2$ and $e^{\log x/x} < 2$, this proves (5.1).

For the second inequality, set $G(\sigma,\tau) = \operatorname{Re}\log M_0(\sigma+i\tau)$. An exact horizontal integration gives

$$G(A,T) - G(0,T) - AL/2 = \tfrac{1}{4}\log(1+A^2/T^2) + \tfrac{1}{2}\log\frac{T^2+(A-1)^2}{T^2+1} + \tfrac{1}{4}\int_0^A \log(1+\sigma^2/T^2)\,d\sigma \ge 0.$$

Every term is nonnegative because $A \ge 2$. On the imaginary axis,

$$\partial_\tau G(0,\tau) = -\frac{\pi}{4} + \frac{1}{2\tau} + \frac{\tau}{1+\tau^2} \le -\frac{\pi}{4} + \frac{3}{2T} \qquad (\tau \ge T).$$

At $A+iT$, the exact formula for α gives $\operatorname{Re}\alpha \ge L/2$ and $0 \le \operatorname{Im}\alpha \le \pi/4$. Thus

$$\operatorname{Re}(\alpha(A+iT)^2) \ge L^2/4 - \pi^2/16.$$

Combining these estimates, including the displacement $T' - T = \pi t/8$, yields

$$e^{t\pi^2/64}\frac{|M_0(iT')|}{|M_t(s_+)|} \le Q^{-A/2}\exp\left(-\frac{tL^2}{16} + \frac{3\pi t}{8x}\right). \tag{5.3}$$

The exponent is nonpositive because $L > 1 > 6\pi/x$ on the stated domain. This proves (5.2). □

6 The complete off-axis budget

Proof of Theorem 1.2. Apply (2.7) with the common cutoff $N = \lfloor\sqrt{Q + t/16}\rfloor$. Lemma 3.1 and (3.5) bound the dominant sum's excess by 95/147. For the opposite sum, use

$$N^{(y+1)/2} \le (Q + t/16)^{(y+1)/4} < 2Q^{(y+1)/4},$$

where $(1 + t/(16Q))^{(y+1)/4} < (33/32)^2 < 2$. Together with Lemma 5.1 this gives

$$\left|\frac{M_t(s_-)}{M_t(s_+)}\right| \sum_{n=1}^{N} n^{(y-1)/2} W_n(s_-) < 4Q^{(1-y)/4} \le 4Q^{-1/2} < \frac{11}{1000}. \tag{6.1}$$

The last step uses $4\pi < 13$ and $16 \cdot 13/(2 \cdot 10^6) < (11/1000)^2$.

The two ordinary approximation errors, normalized by $|M_t(s_+)|$, total less than

$$\frac{\zeta(2)+11/1000}{50000}<\frac{1}{25000}.$$

Lemma 4.1 applies because $T=x/2\ge 10^6$. By (5.2), the two normalized remainder terms total at most

$$2Q^{-(1+y)/4}\le 2/Q<26/x.$$

The triangle inequality now yields

$$\begin{aligned}\left|\frac{H_t(x+iy)}{M_t(s_+)}-1\right| &< \frac{95}{147}+\frac{11}{1000}+\frac{1}{25000}+\frac{26}{2\cdot 10^6}\\ &= \frac{96624791}{147000000}<\frac{33}{50}.\end{aligned}$$

All constants are uniform in t,x,y on the stated domain. □

7 From the comparison to a local count

The following implication is useful independently of the particular cutoff obtained in Theorem 1.2.

Lemma 7.1 (Comparison-to-count extraction)**.** *Fix $0<t\le 1/2$ and $X\ge 200$, and put*

$$U=X+\tfrac12,\qquad w=U+\tfrac72 i,\qquad F_t=H_t/B_t.$$

If $\sup_{|z-w|\le 1/2}|F_t(z)-1|\le 7/10$, then

$$K_t(X)\le \frac{82123}{72000}\log(X+2)+\frac{574}{27}.$$

Proof. The disk has imaginary parts in $[3,4]$, and its image under $s=(1-iz)/2$ has real part at least 2. Therefore B_t is nonzero and F_t is holomorphic on a neighborhood of the closed disk. Cauchy's estimate gives

$$\left|\frac{F_t'}{F_t}(w)\right|\le\frac{7/10}{(1/2)(3/10)}=\frac{14}{3}. \tag{7.1}$$

At $s=(1-iw)/2=9/4-iU/2$, with $L_U=\log(U/(4\pi))$, the exact formula (2.2) gives

$$\begin{aligned}|\alpha(s)| &\le \frac{L_U}{2}+\frac{\pi}{4}+\frac{3}{U}+\frac{81}{16U^2}<\frac{L_U}{2}+1<\frac12\log(X+2),\\ |\alpha'(s)| &\le \frac1U+\frac{6}{U^2}\le\frac{103}{20000}<\frac{3}{500}.\end{aligned}$$

For the last logarithmic comparison, use $4\pi>e^2$ and $U<X+2$. Holding time fixed, the chain rule yields

$$\frac{B_t'}{B_t}(w)=-\frac{i}{2}\alpha(s)\left(1+\frac{t}{2}\alpha'(s)\right),\qquad \left|\frac{B_t'}{B_t}(w)\right|\le\frac{2003}{8000}\log(X+2). \tag{7.2}$$

Evenness, order one and $H_t(0)>0$ give the paired Hadamard identity [1, (105)]. Its imaginary part is the positive scalar sum

$$D_t(U):=-\operatorname{Im}\frac{H_t'}{H_t}(w)=\sum_\rho\frac{7/2-\operatorname{Im}\rho}{(U-\operatorname{Re}\rho)^2+(7/2-\operatorname{Im}\rho)^2}. \tag{7.3}$$

Multiplicity is retained. The scalar sum converges absolutely by the strip bound and order-one zero growth, independently of the conclusion being proved.

For a zero in the unit window set $h = U - \operatorname{Re}\rho$ and $v = 7/2 - \operatorname{Im}\rho$. Then $h^2 \le 1/4$ and $5/2 \le v \le 9/2$, so

$$\frac{v}{v^2+h^2} \ge \frac{9}{41}.$$

For instance, $41v - 9v^2 - 9/4 = 9(9/2 - v)(v - 1/18) \ge 0$. Consequently $K_t(X) \le (41/9)D_t(U)$. Combining (7.1)–(7.3) proves

$$K_t(X) \le \frac{41}{9}\left(\frac{2003}{8000}\log(X+2) + \frac{14}{3}\right),$$

as required. The horizontal distance bound includes both walls, so it also bounds the specified half-open window without a nonvanishing condition on either vertical boundary. □

Proof of Theorem 1.1. For $X \ge 2\cdot 10^6$, the entire disk in Lemma 7.1 lies in the domain of Theorem 1.2. Its comparison hypothesis follows from (1.6). The final rounding uses $82123/72000 < 6/5$, $574/27 < 22$ and $\log(X+2) > 0$. □

8 Scope and limitations

This is an explicit upper count on a restricted high-height domain. An upper count alone supplies neither a lower population, an exact interior integer count, nor a lower bound on a derivative at a zero. The theorem is compatible with multiple zeros and collisions. The kernel argument applies to any function meeting its Hadamard and strip hypotheses; the complete theta expansion is the additional input used to obtain the particular numerical comparison. Positivity, evenness or reflection of a modified source does not supply that expansion.

In particular, the Euler-factor, weighted-square, reflected and positive-primitive control families recorded in the research repository do not challenge a multiplicity-count upper bound merely by colliding. No untested control is claimed to fail (1.6) on a particular box. No noncollision conclusion is drawn.

The lower-height gap is stated rather than filled by compactness. The local argument does not provide an evaluated cumulative estimate for $N_t(X)$. If a later application uses Polymath's cumulative comparison function, the normalization to retain is

$$g(X,t) = \frac{X}{4\pi}\log\frac{X}{4\pi} - \frac{X}{4\pi} + \frac{11}{8} + \frac{t}{16}\log\frac{X}{4\pi}.$$

The heat drift and the constant $11/8$ should not be replaced by a time-zero or averaged convention. A conductor-dependent version for Dirichlet families also requires additional estimates.

The natural next analytic task is to lower the cutoff or cover $200 \le X < 2\cdot 10^6$. A separate application must specify its kernel, domain and numerical tolerance before a gain can be assessed. The present draft makes no assertion that the bound improves an existing certificate.

A Source normalization and reproducibility

The source statements actually used are identified in Section 2. In the proof following (99) of [1], the displayed logarithmic contribution to $\operatorname{Re} s_*$ uses $t/2$, while its definitions (9), (17) and its explicit

estimate (21) give $t/4$. We avoid that display and use the exact identity (3.4). This observation does not dispute the source's time-uniform asymptotic theorem.

In the remainder calculation, the envelope in Section 4 is derived directly from the explicit Stirling and Taylor errors. It uses $u^2 + 1$, rather than copying the source's compressed constant immediately before (60). The term E_a keeps the fixed-index remainder when its index is selected by height. Both choices are conservative and enter the stated numerical budget.

Two standard-library Python programs accompany the research records:

Program in `../numerics/`	Scope
`check_local_count_reduction.py`	Kernel and extraction constants
`check_off_axis_comparison.py`	Off-axis finite constants and budget

The second uses rational Taylor enclosures and checks the Gaussian moment allowances, the two initial tail coefficients, the decreasing tail recurrence, the fixed-index allowance and the final rational budget. Its run record is `../numerics/OFF_AXIS_COMPARISON_CHECKS_20261017.json`; the first record is `../numerics/LOCAL_COUNT_REDUCTION_CHECKS_20261017.json`. The source notes and their hashes remain identified in those records.

The ordinary run of each program exits zero. The first program with `--kernel-floor9/40` exits one because its requested floor exceeds the endpoint value 9/41. The second with `--comparison-ceiling13/20` exits one because that requested ceiling is smaller than the current sufficient budget. Neither injection is a numerical counterexample to H_t. The finite checks do not formally verify the imported theorems, contour shifts or infinite-domain analytic arguments. They constitute arithmetic authoring support, rather than independent mathematical review.

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References

[1] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Research in the Mathematical Sciences **6**, article 31 (2019). doi:10.1007/s40687-019-0193-1. The source numbering used here follows arXiv:1904.12438v2, 4 August 2019.